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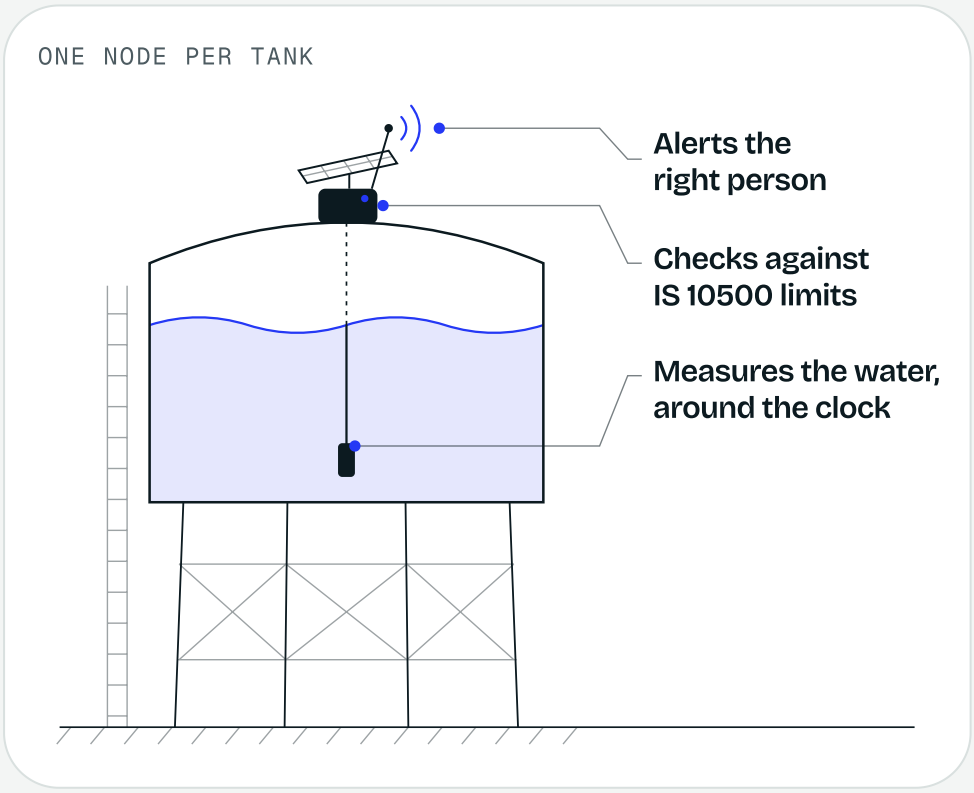
Measured, not *assumed*.

Why India's village overhead tanks need continuous water-quality monitoring, and how Aqua Insight puts it into practice.

BY
Team WAL & AL

VERSION
1.0 · November 2026

READING TIME
About 25 minutes



India has built the taps. Whether the water is safe is still mostly assumed.

Since August 2019, the share of rural Indian homes with a tap has risen from 16.7% to 82.3%. The Jal Jeevan Mission's promise is not only a tap: it is water of the quality set by IS 10500, supplied regularly and over the long term.[1] Quality, unlike a tap, cannot be inspected once and ticked off.

Water can change between the source and the glass, and most of all where it waits: in the overhead tanks that feed village pipe networks. Today it is checked by laboratory samples and field test kits. Those tests are essential, and their scale has grown fast. But each one is a snapshot, and between snapshots the quality of the water is assumed.

The evidence says the assumption often fails. In the Mission's own 2024 assessment of villages that report a tap in every home, 24% of household water samples failed the microbiological test.[2] In its 2022 assessment, residual chlorine, the disinfectant that protects water on its way to the tap, was found within permissible limits in only 24% of household samples; 72% had none at all.[3]

This paper makes the case for monitoring a few operational parameters continuously at every village overhead tank: free residual chlorine, pH, turbidity and total dissolved solids. Each reading should be checked against IS 10500 as it arrives and shared with the people who run the water and the people who drink it. Sections 01–04 set out the evidence, sections 05–09 describe Aqua Insight, the system we are building to do this, and section 10 is plain about what sensors cannot do.

82.3%

of rural households had a tap connection on 31 August 2026, up from 16.7% in August 2019.

JAL JEEVAN MISSION [4]

24%

of household tap-water samples failed the microbiological test in Har Ghar Jal villages, 2024.

FUNCTIONALITY ASSESSMENT [2]

2.04 lakh

diarrhoea deaths in India in 2019 were attributable to unsafe drinking water.

WHO [5]

2.1 billion

people worldwide still lacked safely managed drinking water in 2024.

WHO/UNICEF JMP [6]

IN THIS PAPER

01 From access to assurance
02 Where quality changes
03 What is at stake
04 The gap between tests
05 Aqua Insight
06 How it works
07 The Water Quality Index
08 One view, every role
09 Security and standards
10 What sensors don’t replace
11 Status and roadmap
12 Conclusion

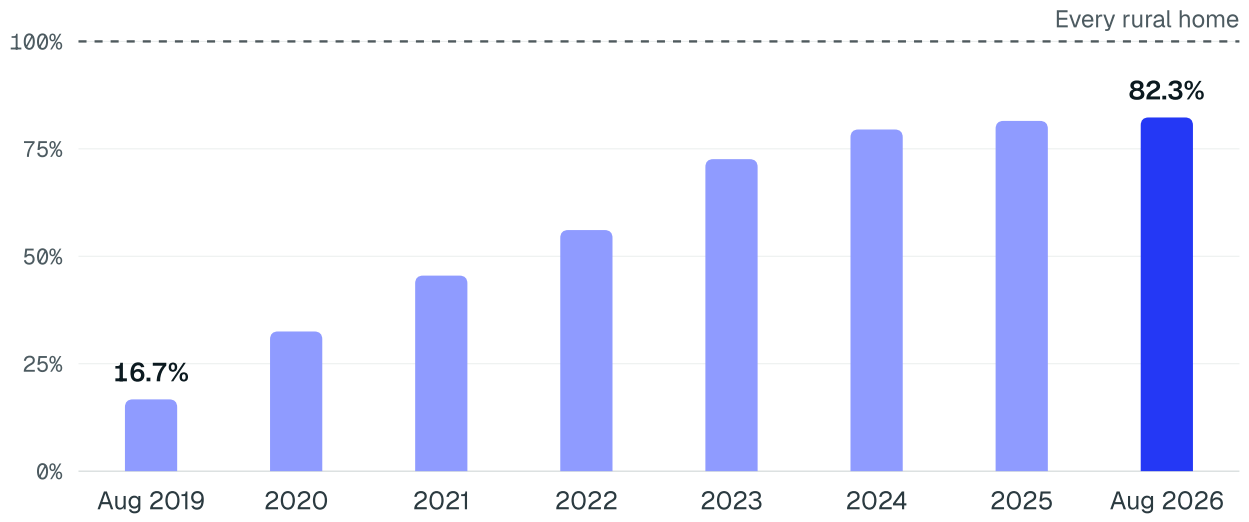
In seven years, rural India went from one home in six with a tap to more than four in five.

When the Jal Jeevan Mission (Har Ghar Jal) began on 15 August 2019, 3.23 crore rural households, 16.7% of the total, had a tap connection. By 31 August 2026, the Mission’s information system recorded 15.94 crore of 19.36 crore rural households, or 82.34%.[7][4] That is about 12.7 crore more homes with a tap in seven years.

FIG. 1

Rural households with a tap connection under the Jal Jeevan Mission, 2019–2026

Share of all rural households: at launch (August 2019), on 31 December of each year, and the latest figure (31 August 2026).



Source: National Jal Jeevan Mission, *Jal Jeevan Samvad* newsletters, December 2020 to August 2026 (snapshots of the Mission's IMIS) [8][4]. The launch figure is 3,23,62,838 households (16.7%) [7]. Percentages are as printed; the total number of rural households changes slightly between snapshots.

The promise includes quality

The Mission does not count a tap as enough. Its guidelines define a functional household tap connection as one that provides water "in adequate quantity, i.e. at least 55 lpcd, of prescribed quality, i.e. BIS:10500 standard, on regular basis, i.e. continuous supply in long-term".[1]

Quantity and regularity can be checked by watching the supply. Quality is harder: it is invisible, it can change in hours, and it is measured today by taking samples to a laboratory or testing them with a kit. As coverage approaches every home, the question for the next phase moves from *is there a tap?* to *is the water safe, today, at this tap?*

> "What cannot be measured cannot be managed. … it is necessary to have data on water supplied every day, the quality of water supplied as well as depth of water in bore wells."
>
> JAL JEEVAN MISSION, OPERATIONAL GUIDELINES, DECEMBER 2019, §11.1 [1]

Water that leaves the source safe does not always arrive safe.

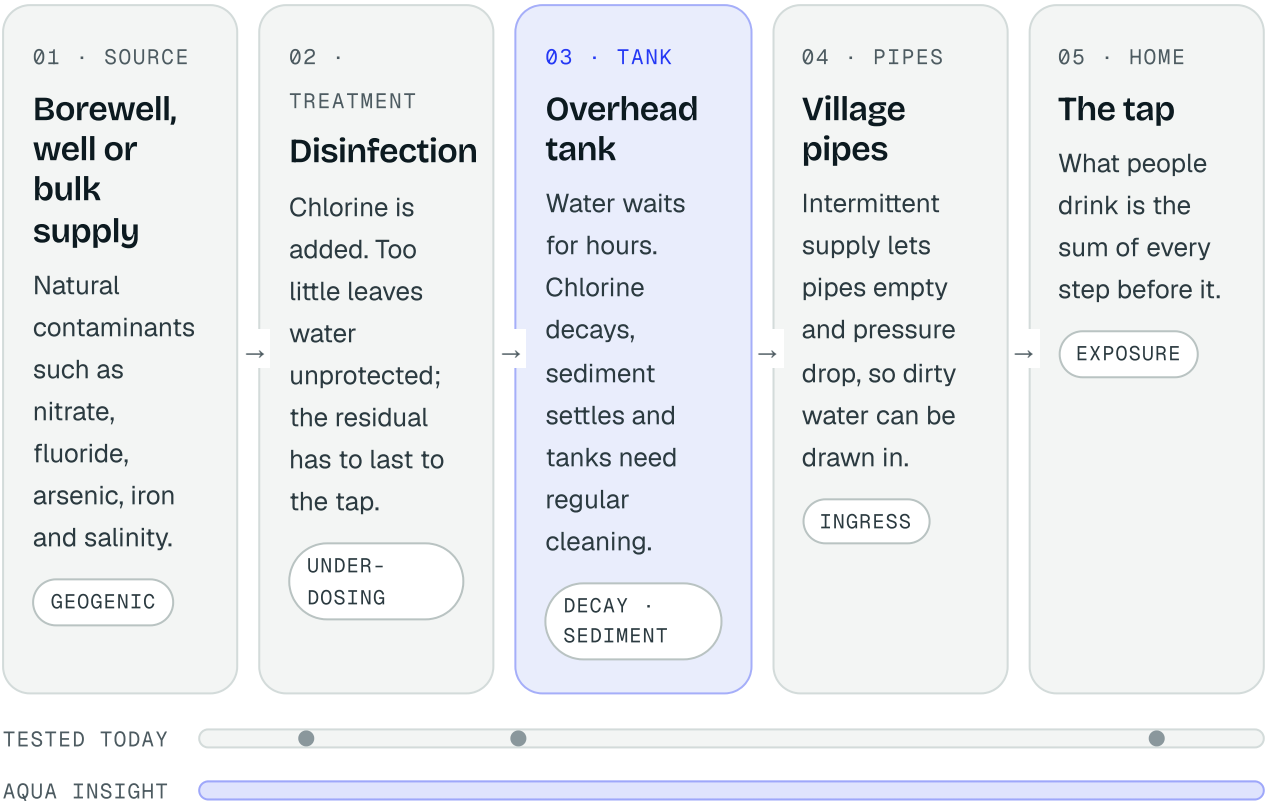
In a typical village scheme, water is pumped from a borewell, well or bulk supply into an overhead tank, and flows from there by gravity through the village pipes to each home. The Mission's guidelines list the overhead tank as part of the in-village infrastructure, and ask for the supply from each one to be measured with sensors.[1] In the Mission's 2020–21 functionality survey, 55.5% of villages reported an overhead tank in their piped water scheme.[9]

Between the treatment point and the tap, three things can go wrong, and each leaves a trace in the water. WHO notes that outbreaks in distribution systems "have been linked to cross-connections, contamination during storage, low water pressure and intermittent supply".[10]

- **The disinfectant runs out.** Chlorine is added to kill germs, and enough of it has to survive all the way to the tap. WHO advises that "a chlorine residual should be maintained throughout the distribution system" and that free chlorine at the point of delivery should be at least 0.2 mg/l.[10] The government's 2024 water supply manual is blunt: "absence of residual chlorine could indicate potential presence of heavy contamination."[11]
- **Stored water deteriorates.** Water quality in a reservoir "may also deteriorate due to excessively prolonged periods of stagnant conditions".[11] Unprotected tanks "are at risk of faecal contamination from birds and vermin".[12] A February 2026 advisory for urban water systems asks for reservoirs and overhead tanks to be cleaned "at least twice a year".[13]
- **Intermittent supply lets dirt in.** When pipes run empty between supply hours, pressure drops and contaminated water can be drawn in through joints and cracks. "It is now well established that intermittent water supply leads to health risks for users," the same manual states.[14] WHO calls intermittent supplies "frequently associated with contamination".[10]

FIG. 2

From source to tap: where water quality can change, and where it is checked today



Schematic. Dots mark the usual sampling points under the Mission's framework: drinking-water sources, treatment plants and tail-end taps, tested at set intervals [15]. Aqua Insight adds a continuous record at the overhead tank, the point every home on the scheme shares.

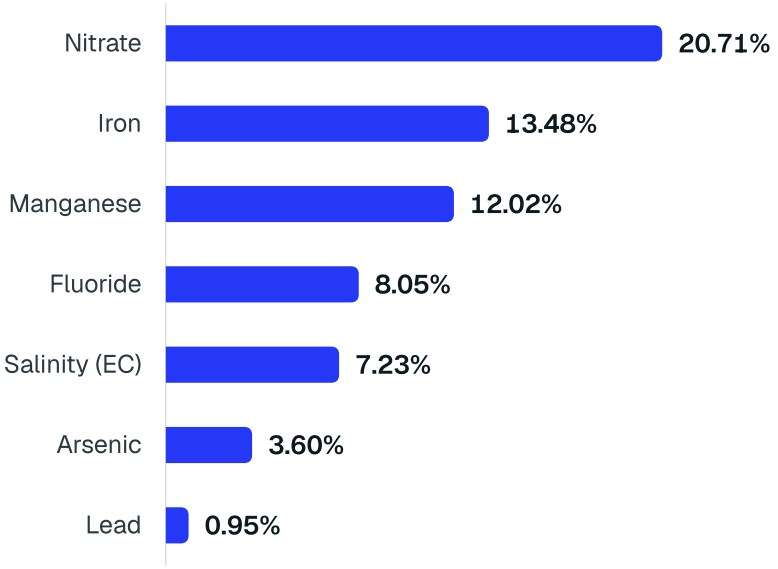
The source itself is not always safe

Many rural schemes draw on groundwater, and in many districts it carries natural contaminants that no amount of storage management will remove. In the Central Ground Water Board's 2024 sampling, one in five samples exceeded the BIS limit for nitrate, and one in twelve exceeded it for fluoride.[16]

FIG. 3

Groundwater samples above the BIS limit, by parameter, India (pre-monsoon 2024)

Share of samples tested for each parameter.



Source: Central Ground Water Board, state-wise tables of the Annual Ground Water Quality Report 2025 (pre-monsoon 2024 samples) [16]. Iron, manganese, arsenic and lead were tested in a smaller set of samples. These are groundwater samples from CGWB's monitoring network, not samples of supplied drinking water.

These contaminants need laboratory tests and, where they are found, treatment or a different source. Continuous sensors at the tank do not detect them (section 10). As reported by States, access to potable water has been ensured in every arsenic- and fluoride-affected habitation; in January 2025 the last 568 of them were served by interim community water purification plants.[17]

03 — WHY IT MATTERS

WHAT IS AT STAKE

Unsafe water is still one of India's largest preventable causes of illness and death.

WHO's latest burden-of-disease estimate, for 2019, attributes 4,97,738 deaths in India to unsafe drinking water, sanitation and hygiene. That is 36.4 deaths per 1,00,000 people. Unsafe drinking water alone accounted for 2,03,863 deaths from diarrhoea.[5] Worldwide, at least 1.7 billion people used a drinking-water source contaminated with faeces in 2022.[18]

Routine reporting shows the illness behind those deaths. In 2022, States reported nearly 56 lakh cases of acute diarrhoeal disease and more than 12.3 lakh cases of typhoid (provisional figures).[19] In the National Family Health Survey of 2019–21, 7.3% of children under five had had diarrhoea in the two weeks before the survey, down from 9.2% in 2015–16.[20]

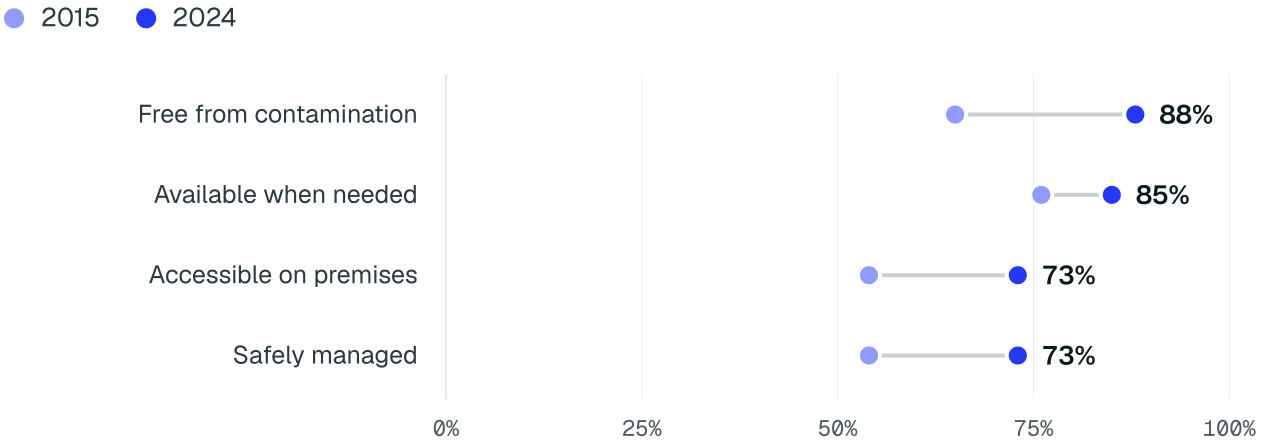
Measuring quality is what made India's progress visible

The WHO/UNICEF Joint Monitoring Programme (JMP) counts drinking water as "safely managed" only when it comes from an improved source that is on premises, available when needed and free from faecal and priority chemical contamination.[21] Until 2025 it had no national estimate for India. Its 2025 report was the first, "due to newly compiled data on drinking-water quality in urban areas".[6] Water-quality data is what made the national figure possible.

FIG. 4

Rural India against the three tests of safely managed drinking water, 2015 and 2024

Share of the rural population. Safely managed is the lowest of the three.



Source: WHO/UNICEF JMP, *Progress on household drinking water, sanitation and hygiene 2000–2024* (2025), Annex 3, p. 123 [6]. For India as a whole in 2024: 76% safely managed and 96% with at least a basic service.

Even on the JMP's measure, about one rural Indian in eight in 2024 did not have drinking water from an improved source free from contamination. Tests at the tap tell a sharper story: in the Mission's 2024 assessment of Har Ghar Jal villages, 24% of household samples failed the microbiological test.[2]

04 — THE GAP

BETWEEN TESTS

India tests more water than ever. Each test is still a snapshot.

The Mission has built a very large water-testing network. In January 2025 there were 2,162 water-testing laboratories, 1,572 of them accredited or recognised by NABL.[17] In 2025–26, up to 21 October 2025, 2,843 laboratories tested 38.78 lakh samples.[22] In 2024–25, up to 15 January 2025, laboratories tested 62 lakh samples and found 3.18 lakh contaminated, and 83 lakh more were tested with field test kits. 24.8 lakh women have been trained to use the kits.[17]

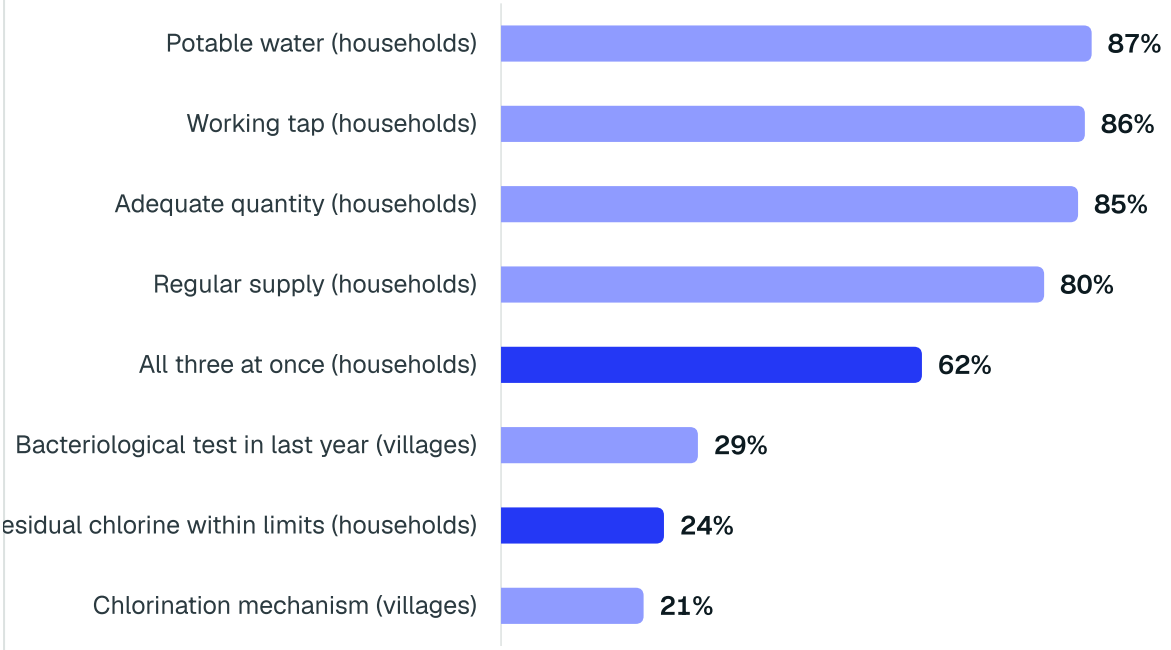
The testing schedule is set by the Mission's 2021 framework. Laboratories test every drinking-water source "at least twice a year, i.e., before and after the monsoon for bacteriological parameters and at least once in a year for chemical parameters". In every village, five trained people, preferably women, test the sources and two tail-end taps with field test kits at least once a month.[15] The framework itself notes that a kit "gives an indicative result".

The Mission's own surveys show what a monthly rhythm can miss. In 2022, 87% of households received water that passed laboratory tests. Yet residual chlorine was within permissible limits in only 24% of household samples, and only 21% of villages had a mechanism for chlorination.[3]

FIG. 5

Household tap connections against JJM's functionality tests, 2022

Share of surveyed households or villages, Functionality Assessment 2022 (February–April 2022).



Source: National Jal Jeevan Mission, *Functionality assessment of household tap connection – 2022*, national factsheet, p. 17 [3]. The 2024 round reports 76% of household samples passing the microbiological test and chlorination available in 70.3% of villages, but covers only Har Ghar Jal villages and says it is not directly comparable with earlier rounds [2].

What a snapshot misses

A sample shows the water at one moment. Free chlorine can drop quickly: when contamination gets in, when water stands too long in a tank, or when supply is interrupted. WHO notes that "a sudden disappearance of an otherwise stable residual can indicate ingress of contamination".[10] If the drop comes and goes between two samples, a household can drink unprotected water for days without any test recording it.

FIG. 6

Two samples a month apart can both pass while the water fails in between

Schematic: the shape of a typical event, not measured data.

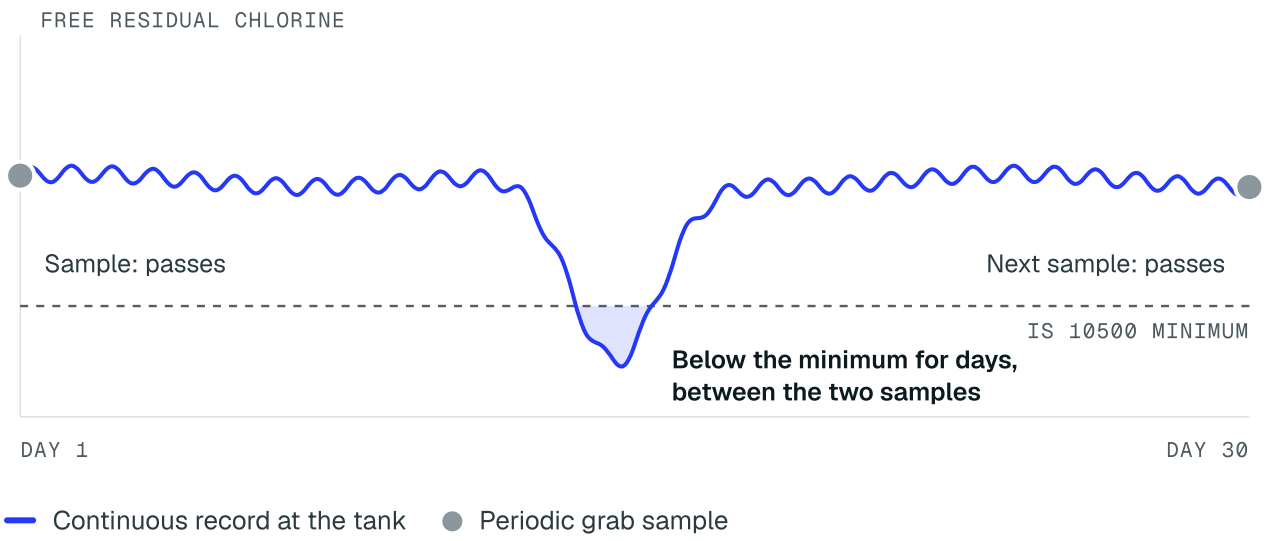


Illustration only. It shows why sampling frequency matters; it does not report readings from any tank.

The official direction is already set

None of this is a new idea. The Mission's 2019 guidelines ask for the supply "from each ESR [elevated storage reservoir] to be measured using modern sensors based IoT solutions", and for public health engineering departments to "employ water quality sensors and IoT based monitoring systems".[1] Its 2021 framework calls for "technological innovations such as sensor-based IoTs which provide automatic data on water supply quantity, quality and regularity", starting with pilot projects to be "demonstrated, and later be planned for scaling it up".[15]

The government's 2024 operations manual goes further: "Wherever online water quality parameters are monitored, the laboratory water quality test may not be required for such parameters."[11] WHO explains why speed matters: laboratory tests for germs "are often of limited use for operational monitoring, because the time taken to process and analyse water samples does not allow operational adjustments to be made prior to supply".[10]

THE OPPORTUNITY

Laboratories and field-test kits tell us what is in the water. Continuous sensors at the tank tell us when it changed. Together they turn periodic checks into a record that operators, officials and residents can act on, and trust.

One node on every tank, one shared view of the water.

Aqua Insight is the drinking-water system Team WAL & AL is building for village overhead tanks. A solar-powered node on the tank measures the water around the clock. Every reading is checked against IS 10500 as it arrives, and the result reaches the people who run the tank, the officials who oversee it and the residents who drink from it.

What the node measures

The node measures four parameters continuously: free residual chlorine, pH, turbidity and total dissolved solids (TDS). Water level is planned as a fifth reading, so the same view shows supply as well as quality. We chose these four for two reasons. They can be measured reliably by sensors that sit in the water. And between laboratory tests, they answer the questions that matter most: is the water still disinfected, has something disturbed or entered it, and has its mineral content changed?

TABLE 1

The four parameters Aqua Insight measures continuously, with their limits in IS 10500:2012

PARAMETER	ACCEPTABLE LIMIT	PERMISSIBLE LIMIT, NO ALTERNATE SOURCE	WHAT IT TELLS YOU BETWEEN TESTS
Free residual chlorine	At least 0.2 mg/l	1 mg/l	Whether the water is still protected. The standard tests it “at consumer end”, and asks for at least 0.5 mg/l where protection against viral infection is required. A sudden loss can signal contamination.
pH	6.5–8.5	No relaxation	Whether chlorine can work. WHO advises a pH preferably below 8 for effective chlorination; lower-pH water (about pH 7 or less) is more likely to be corrosive.
Turbidity	1 NTU	5 NTU	Whether particles are present that hinder disinfection. WHO advises keeping it below 1 NTU. A rise can come from sediment or biofilm stirred up in a tank, or from dirt getting in.
Total dissolved solids	500 mg/l	2,000 mg/l	The salts and minerals in the water. They affect taste, and a sudden change shows that the water itself has changed, for example a different source.

Limits: BIS, IS 10500:2012, Tables 1 and 2 [23]. The chlorine requirement applies “only when water is chlorinated”. Guidance on pH and turbidity: WHO, *Guidelines for drinking-water quality*, 2022 [10]; turbidity from sediments and biofilms in storage tanks: WHO, *Water safety in buildings* [12].

What it does with each reading

01 **Checks it against IS 10500** as it arrives, and flags a breach in real time.

02 **Warns before a limit is crossed.** A trend fitted to recent readings projects the next few hours, so a falling chlorine level raises a warning while there is still time to act.

03 **Alerts the right person.** Alerts go to the people responsible for that tank, who acknowledge and resolve them in the app. Each action is recorded, so it is clear who acted and when.

04 **Scores the water.** A 0–100 Water Quality Index for every tank, village and district (section 07).

05 **Watches its own sensors.** It flags readings that flat-line and calibration that is due, so a faulty sensor is not mistaken for safe water.

06 **Closes the loop.** Rules can turn readings into commands sent back to the device, within limits that operators set.

07 **Reports compliance.** IS 10500 reports by village, exported as PDF.

08 **Answers the resident's question.** An "Is the water safe?" card gives a plain answer, checked against IS 10500.

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06 — HOW IT WORKS SENSE · CONNECT · UNDERSTAND · ACT

From the tank to a decision, in one loop.

Aqua Insight runs on the same four-step method behind every WAL & AL system. We build and run every layer ourselves, from the node's firmware to the apps, so the system behaves as one product and a problem in any layer is ours to fix.

FIG. 7

The Aqua Insight loop: every reading travels from the tank to a decision, and decisions can travel back.

01 · SENSE	02 · CONNECT	03 · UNDERSTAND	04 · ACT
Measure at the tank	**Bring every reading home**	**Turn data into a signal**	**Close the loop**
A solar-powered node with sensors in the water reads the four parameters around the clock.	Each reading travels over an encrypted link from an authenticated device to a cloud hosted in India.	Readings enter a durable queue, are checked against safe limits and are kept as a time-series history.	Alerts, dashboards and reports reach each role, and commands can go back to the device.
Solar and battery power Multi-parameter sensors Actuators for commands	Encrypted in transit One identity per device Hosted in India	Nothing dropped under load Checks against IS 10500 Trends and the WQI	Alerts to the right person Phone and web apps Commands back to devices

Method shared by every WAL & AL system. Specific components are not named in this paper.

Design choices

- **Built for the tank, not the lab.** The node runs on solar power and a battery, so it does not depend on mains power at the tank.
- **Hosted in India.** Readings, accounts and logs are kept in Indian data centres.
- **One app for every role, on phone and web.** Owners, operators and officials get the same features on both, from one codebase, so field staff and offices see the same thing.
- **Nothing lost in transit.** Readings pass through a durable queue before they are processed, so a burst of traffic or a slow step does not drop data.

One score from 0 to 100, without hiding a breach.

Four readings per tank, across hundreds of tanks, are too many numbers for a busy official. The Aqua Insight Water Quality Index (WQI) turns each tank's readings into one score from 0 to 100, which also averages up to a village and a district.

01 **Score each parameter.** Inside its safe band, a parameter scores 85–100, highest near the middle of the band. For turbidity and TDS, lower is better, so low values are not penalised. Across the caution band just beyond the limit, the score falls from 85 to 50. Beyond that, it falls from 50 towards 0.

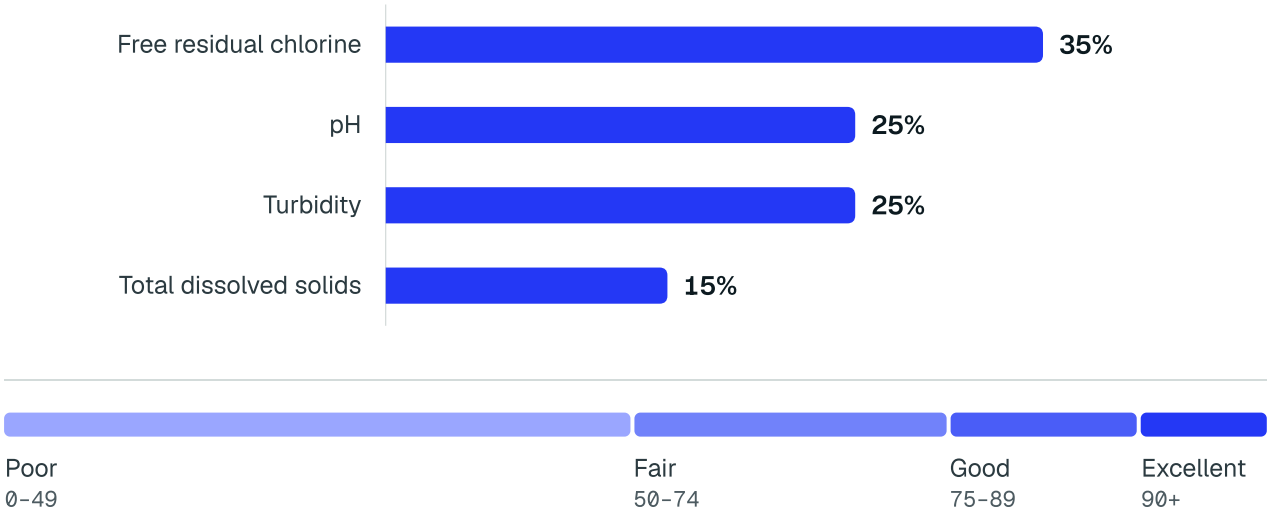
02 **Weight the scores.** The weights favour the parameters most directly tied to health: free residual chlorine 35%, pH 25%, turbidity 25% and TDS 15% (Fig. 8).

03 **Grade the result.** 90 and above is Excellent, 75–89 Good, 50–74 Fair, and below 50 Poor.

FIG. 8

How much each parameter counts towards the Water Quality Index.

Weights in version 1.0 of the index. Disinfection carries the most weight.



Source: Aqua Insight WQI, version 1.0. The WQI is our own index. It is not a BIS or government index.

THE INDEX NEVER OVERRIDES A LIMIT

The WQI summarises; it does not decide. Each reading is still checked against its own IS 10500 limit, and a single parameter outside its limit raises its own alert, whatever the overall score.

The bands and weights in version 1.0 are a starting point. We will check them against laboratory results in field trials and record any change in later versions of this paper.

08 — PEOPLE

ONE VIEW, EVERY ROLE

The same live record for everyone responsible for the water.

Village water is everyone’s business and often no one’s job. Aqua Insight gives each person a view scoped to what they are responsible for, built on one shared record. A problem is then visible to the people who can fix it and to the people who oversee them.

TABLE 2

Roles in Aqua Insight. Every role gets the same features on phone and web.

ROLE	WHO	SEES	CAN DO
Resident	People who drink the water	Their tank’s status	Check “Is the water safe?” in plain words
Owner	The tank’s owner, such as a gram panchayat, a community or an institution	Their own tanks	Acknowledge alerts, see history and reports
Operator	Field staff who run, clean and repair tanks	Assigned tanks	Acknowledge and resolve alerts, log work, act on the device, see reports
Authority	District water officials	Their district	Oversight of every tank, compliance reports, fleet view (read-only)

09 — TRUST

SECURITY, PRIVACY AND STANDARDS

A system that vouches for water has to be trustworthy itself.

If the data can be faked, lost or leaked, the “safe” card means nothing. Aqua Insight is designed to the Indian standards and rules that apply to drinking water, personal data and cyber-security, and to the leading open standard for mobile app security.

TABLE 3

The standards and rules Aqua Insight is designed to meet

STANDARD	WHAT IT REQUIRES	HOW AQUA INSIGHT IS DESIGNED TO MEET IT
IS 10500:2012 BIS	Acceptable and permissible limits for drinking water. E. coli “shall not be detectable in any 100 ml sample”.	Every reading is checked against its IS 10500 limit as it arrives. Compliance reports by village.
Digital Personal Data Protection Act, 2023 ACT 22 OF 2023	Notice, and consent that is free, specific and informed. “Reasonable security safeguards to prevent personal data breach.” Breach intimation to the Data Protection Board and each affected person. The DPDP Rules, 2025, notified on 13 November 2025, bring the main duties into force 18 months later.	Only the personal data needed for sign-in and roles. Notice and consent in the apps. Consent records, correction and erasure handled on the server. A breach procedure.
CERT-In Directions, 28 April 2022 IT ACT S. 70B(6)	Report listed cyber incidents to CERT-In within 6 hours. Keep ICT logs for a rolling 180 days within Indian jurisdiction. Synchronise clocks with NIC or NPL time servers.	Logs kept for 180 days in India. Clocks synchronised to national time servers. An incident runbook with 6-hour reporting.
OWASP MASVS v2.1.0 JANUARY 2024	The industry standard for mobile app security, covering storage, cryptography, authentication, network, platform, code, resilience and privacy.	Phone apps designed to MASVS controls, such as encrypted session storage and lockout after repeated failed sign-ins.
UN SDG indicator 6.1.1 SAFE DRINKING WATER	Water from an improved source that is on premises, available when needed and free from faecal and priority chemical contamination.	Continuous evidence on water quality at the point every home on a scheme shares.

Sources: IS 10500 [23]; DPDP Act [24] and Rules [25]; CERT-In Directions [26]; OWASP MASVS [27]; SDG 6.1.1 metadata [21].

Security controls, by design

These controls are part of Aqua Insight’s design, and an independent audit will test them before launch.

- **Devices.** Each node has its own identity and may send only its own readings. Commands to devices are checked on the server, against the sender's role and the tank's limits.
- **Data.** Encrypted in transit and at rest. Data and logs are kept in India.
- **Access.** The same role and tank-scope checks run on every request and every device command, not only in the app's screens. Officials and administrators use multi-factor sign-in.
- **Apps.** Short-lived sessions in encrypted storage on the device, lockout after repeated failed sign-ins, an optional biometric lock, and no keys or credentials inside the app.
- **Assurance.** An independent vulnerability assessment and penetration test by a CERT-In empanelled auditor before launch, then every year.

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10 — LIMITS

WHAT SENSORS DON'T REPLACE

Continuous sensors are not a laboratory.

We think it matters to say clearly what continuous monitoring does not do.

- **They do not detect germs directly.** IS 10500 requires that E. coli "shall not be detectable in any 100 ml sample",[23] and only a microbiological test can show that. Free chlorine, pH and turbidity show whether the conditions for safe water hold. WHO describes chlorine residual monitoring as giving "a rapid indication of problems that will direct measurement of microbial parameters".[10] It tells you when to test; it does not replace the test.
- **They do not cover every contaminant.** Arsenic, fluoride, nitrate, iron, uranium and many other contaminants need laboratory tests (section 02). TDS is not a stand-in for any of them.
- **They need care.** Sensors in water drift and foul, and must be cleaned and calibrated on schedule. Aqua Insight flags calibration that is due and readings that flat-line, but someone still has to do the work.
- **The tank is not every tap.** The node measures the water in the tank. Pipes downstream can still let dirt in, especially when supply is intermittent, and IS 10500 tests free chlorine "at consumer end".[23] Testing at tail-end taps stays essential.

WHAT CONTINUOUS MONITORING ADDS

Timing. It shows when something changed, where, and for how long. Laboratory and field-kit tests can then be sent where they are needed, and a problem can be found within hours rather than at the next scheduled sample.

Where Aqua Insight is today, and what comes next.

Aqua Insight is in development. The node, the cloud platform and the apps are being built and tested together against a simulated tank, ahead of field trials. Nothing in this paper describes a deployed system or reports results from one.

FIG. 9

Four phases, each with a gate that must be passed before the next begins.

PHASE 1 · NOW

Build and bench test

Node, cloud platform and apps built together and tested against a simulated tank and on the bench.

GATE
Readings flow end to end, and every alert reaches the right role.

PHASE 2

Field trial

Nodes on a small number of real tanks, run alongside laboratory tests of the same water.

GATE
Sensor readings agree with laboratory results, and nodes run unattended on solar power.

PHASE 3

Pilot

A cluster of villages with a gram panchayat or district partner, with every role using the system.

GATE
Alerts are acted on in time, and officials use the reports.

PHASE 4

Scale

More tanks, villages and districts on the same platform and method.

GATE
Cost and upkeep per tank that a village or district can sustain.

We are looking for partners for phases 2 and 3: gram panchayats and village water committees, district and state water departments, NGOs and CSR programmes, and research labs that can run laboratory tests alongside the sensors.

India has built the taps. The next task is knowing the water is safe, around the clock.

The Jal Jeevan Mission has changed what rural India can expect from its water supply. The next step is to make sure that what flows from the taps meets the standard the Mission itself set: water of prescribed quality, supplied regularly, over the long term.

Laboratories and field test kits will remain the reference for what is in the water. Continuous monitoring at the overhead tank adds what they cannot: a record of when the water changed, shared at once with the people who can act on it. The Mission's own guidelines already ask for sensors. What remains is to prove them in the field, at a cost and an upkeep that villages can sustain.

That is the work Aqua Insight is built for. We would like to do it with the people who run, oversee and drink village water.

Sources

Official and primary sources only. Figures are quoted as published, with the edition and date we used.

[1] Ministry of Jal Shakti, Department of Drinking Water and Sanitation, National Jal Jeevan Mission. **Operational Guidelines for the Implementation of Jal Jeevan Mission (Har Ghar Jal)**. December 2019. jaljeevanmission.gov.in/sites/default/files/guideline/JJM_Operational_Guidelines.pdf
Definition of a functional household tap connection, p. 5; sensors for each ESR, p. 38; water-quality sensors, §10.5, p. 63; §11.1, p. 64.

[2] National Jal Jeevan Mission. **Functionality Assessment of Household Tap Connection: National Report 2024**. Survey July–October 2024, Har Ghar Jal villages only. jaljeevanmission.gov.in/sites/default/files/2025-12/FHTC_National%20Report%202024.pdf
National factsheet, printed p. 2; water quality, §4.6.1, printed p. 33. The report states it is not directly comparable with earlier rounds.

[3] National Jal Jeevan Mission. **National Report: Functionality assessment of household tap connection – 2022**. Survey February–April 2022. jaljeevanmission.gov.in/sites/default/files/2022-10/national_report_of_functionality_assessment_2022.pdf
Executive summary, p. 12, and national factsheet, p. 17.

[4] National Jal Jeevan Mission. **Jal Jeevan Samvad**. 71st edition, August 2026. jaljeevanmission.gov.in/sites/default/files/2026-10/Jal-Jeevan-Samvad-August-2026.pdf
JJM-IMIS snapshot, printed p. 6: 15,93,64,282 of 19,35,54,073 rural households (82.34%) as on 31 August 2026.

[5] World Health Organization. **Burden of disease attributable to unsafe drinking-water, sanitation and hygiene: 2019 update**. 2023. www.who.int/publications/i/item/9789240075610
India, Annex 6, Tables A6.1 and A6.2.

[6] WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation and Hygiene. **Progress on household drinking water, sanitation and hygiene 2000–2024: special focus on inequalities**. August 2025. washdata.org/reports/jmp-2025-wash-households
Key findings, p. viii; India estimates, Annex 3, pp. 122–123; India data note, p. 3.

[7] National Jal Jeevan Mission. **Jal Jeevan Samvad**. 51st edition, December 2024. jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeevanSamvad-December-2024-en.pdf
JJM-IMIS snapshot, printed p. 5: households with tap water on 15 August 2019, 3,23,62,838; on 31 December 2024, 15,40,21,333 (79.53%).

[8] National Jal Jeevan Mission. **Jal Jeevan Samvad**. December issues of 2020, 2021, 2022, 2023 and 2025 (JJM-IMIS snapshots). jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeevanSamvad-December-2020-en.pdf · jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeevanSamvad-December-2021-en.pdf · jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeevanSamvad-December-2022-en.pdf · jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeevanSamvad-December-2023-en.pdf · jaljeevanmission.gov.in/sites/default/files/2026-01/Jal-Jeevan-Samvad-December-2025.pdf
Coverage on 31 December of each year: 32.54%, 45.51%, 56.10%, 72.55% and 81.49%.

[9] National Jal Jeevan Mission. **Assessment of functionality status of household tap connections in rural areas (2020–2021)**. Final report; survey November–December 2020. jaljeevanmission.gov.in/sites/default/files/2024-04/Functionality-Assessment-National-Report-2020.pdf

[10] World Health Organization. **Guidelines for drinking-water quality: fourth edition incorporating the first and second addenda**. 2022. www.who.int/publications/i/item/9789240045064
Pages quoted: 57, 62, 125, 151, 202 and 245. WHO published a further edition, incorporating a third addendum, on 17 June 2026.

[11] Central Public Health and Environmental Engineering Organisation (CPHEEO), Ministry of Housing and Urban Affairs. **Manual on Water Supply and Treatment Systems (Drink from Tap), Part B: Operation and Maintenance**. Second edition, March 2024. mohua.gov.in/static/uploads/2025/10/1086f3271a4a4ff64a70e79669ee4f2b.pdf
Reservoir water quality, §6.2.3.2, p. 251; chlorine residual, §7.4.17, p. 276; online monitoring, Table 8.3, p. 303.

[12] World Health Organization. **Water safety in buildings**. 2011. www.who.int/publications/i/item/9789241548106
Turbidity from storage tanks, §2.1.2, p. 6; ingress of contamination, §4.6.2, p. 46.

[13] CPHEEO, Ministry of Housing and Urban Affairs. **Advisory on Protection of Water Quality in Distribution System**. AMRUT 2.0, February 2026. mohua.gov.in/static/uploads/2026/03/1f50bd8f27197b40e6e472b5b86f4b8a.pdf
Scheduled cleaning and disinfection of storage facilities, §3.13, p. 18.

[14] Central Public Health and Environmental Engineering Organisation (CPHEEO), Ministry of Housing and Urban Affairs. **Manual on Water Supply and Treatment Systems (Drink from Tap), Part A: Engineering — Planning, Design and Implementation**. Fourth edition, March 2024. mohua.gov.in/static/uploads/2025/10/0eb252b5d7932a03e97fc364f9a503f3.pdf
Intermittent supply, §11.24, p. 677.

[15] Ministry of Jal Shakti, Department of Drinking Water and Sanitation, National Jal Jeevan Mission. **Drinking Water Quality Monitoring & Surveillance Framework**. Released 2 October 2021. jaljeevanmission.gov.in/sites/default/files/guideline/WQMS-Framework.pdf
Sensor-based IoT, p. 6; testing frequency and field test kits, §3.3, p. 16 and §4.3, pp. 20–23.

[16] Central Ground Water Board, Ministry of Jal Shakti. **State-wise details of ground water quality sampling and analysis for various parameters, based on the Annual Ground Water Quality Report 2025**. Pre-monsoon 2024 samples; published December 2025. cgwb.gov.in/cgwbpnm/public/uploads/documents/17648336335318474 33file.pdf
National totals from the state-wise tables.

[17] Department of Drinking Water and Sanitation, Ministry of Jal Shakti. **Annual Report 2024–25**. 2025. www.jalshakti-ddws.gov.in/static/uploads/2025/03/b411cf0455523419696301fe6c08b262.pdf
Quality-affected habitations, §3.4.1, printed p. 69; laboratories and testing (WQMIS snapshot of 15 January 2025), §3.6.2–3.6.3, printed pp. 81–82.

[18] World Health Organization. **Drinking-water**. Fact sheet, 13 September 2023. www.who.int/news-room/fact-sheets/detail/drinking-water

[19] Central Bureau of Health Intelligence, Ministry of Health and Family Welfare. **National Health Profile 2023**. 18th issue, 2024. cbhidghs.mohfw.gov.in/sites/default/files/NHP/NHP-2023-Last-Final.pdf
Tables 3.1.6(B) and 3.1.8(B), 2022 (provisional). Reported cases from routine surveillance, not incidence.

[20] International Institute for Population Sciences (IIPS) and ICF. **National Family Health Survey (NFHS-5), 2019–21: India and State Factsheet Compendium**. Ministry of Health and Family Welfare. dhsprogram.com/pubs/pdf/OF43/NFHS-5_India_and_State_Factsheet_Compendium_Phase-II.pdf
India key indicators, indicator 69, printed p. 4. Also published by IIPS at rchiips.org.

[21] United Nations Statistics Division (custodians WHO and UNICEF). **SDG indicator metadata: indicator 6.1.1**. Last updated 27 September 2024. unstats.un.org/sdgs/metadata/files/Metadata-06-01-01.pdf

[22] Press Information Bureau, Ministry of Jal Shakti. **Jal Jeevan Mission (backgrounder)**. 26 October 2025. static.pib.gov.in/WriteReadData/specificdocs/documents/2025/oct/doc20251026676401.pdf
“Quality Assurance and Monitoring”, p. 4: 2,843 laboratories tested 38.78 lakh samples in 2025–26, as on 21 October 2025.

[23] Bureau of Indian Standards. **IS 10500:2012 Drinking Water — Specification (Second Revision)**. May 2012, with amendments. law.resource.org/pub/in/bis/S06/is.10500.2012.pdf
Tables 1, 2 and 6. Later amendments changed the limits for iron (1.0 mg/l) and arsenic (0.01 mg/l, no relaxation), as reproduced in the CPHEEO 2024 manual; the four parameters in Table 1 of this paper are unchanged.

[24] Ministry of Law and Justice. **The Digital Personal Data Protection Act, 2023 (No. 22 of 2023)**. Gazette of India, 11 August 2023. www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf
Sections 5, 6, 8(5) and 8(6).

[25] Ministry of Electronics and Information Technology. **Digital Personal Data Protection Rules, 2025**. G.S.R. 846(E), 13 November 2025. www.meity.gov.in/static/uploads/2025/11/53450e6e5dc0bfa85ebd78686cadad39.pdf
Rule 1: most rules come into force eighteen months after publication.

[26] Indian Computer Emergency Response Team (CERT-In), Ministry of Electronics and Information Technology. **Directions under sub-section (6) of section 70B of the Information Technology Act, 2000**. No. 20(3)/2022-CERT-In, 28 April 2022. www.cert-in.org.in/PDF/CERT-In_Directions_70B_28.04.2022.pdf
Directions (i), (ii) and (iv).

[27] OWASP Mobile Application Security project. **Mobile Application Security Verification Standard (MASVS)**. Version 2.1.0, January 2024. mas.owasp.org/MASVS/

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